Route Pricing: The Two-Floor Theorem

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November 3, 2026

Abstract

A route has two floors, not one. The single-drop pricing rule gives the *dedicated* floor—what you must charge if the drop travels alone. On a shared route that floor is the wrong number, because the shared costs are being divided. This paper derives the route-level floor, the allocation rule that divides a route's cost among its drops, and the route test that decides whether the route as a whole is provable. We state four allocation rules in order of preference, work through the CBD → Thika example step by step, and derive the four operational rules that follow. The payoff of consolidation is made explicit: every customer pays less than a dedicated trip would cost, and the worst-addressed drop bears the largest share because it consumes the most driver time.

1 The two floors

The single-drop pricing rule gives the dedicated floor: what you must charge if the drop travels alone. On a shared route, that floor is the wrong number, because the shared costs are being divided. A drop on a route can be priced below its dedicated floor and still be profitable for the route—this is exactly why consolidation lets you serve customers who could not afford a dedicated trip.

So route pricing has two questions:

1. Is the route as a whole provable? (route-level test)
2. How is the route's cost divided among its drops? (allocation rule)

2 The route pricing equation

For a route R with drops $j \in R$, the route floor (total) is:

$$F_R = \frac{c_0 + \Phi C_p + c_{\text{km}} D_R}{\bar{p} \cdot \bar{A}} \tag{1}$$

This is the total net revenue the route must earn to cover its fixed costs, principal burden, and distance.

The allocated price for drop j is:

$$n_j = \underbrace{\frac{c_{\text{var},j} + (1 - \hat{p}_j) c_r + \mathbb{E}[\text{loss}_j]}{\hat{p}_j \cdot A(a_j)}}_{\text{standalone component}} + \underbrace{\frac{s_j \cdot (c_0 + \Phi C_p + c_{\text{km}} D_R)}{\hat{p}_j \cdot A(a_j)}}_{\text{allocated shared component}} \tag{2}$$

where s_j is the drop's share of the shared cost, and $\sum_j s_j = 1$.

The route test:

$$\sum_{j \in R} n_j \cdot \hat{p}_j A(a_j) \;\geq\; c_0 + \Phi C_p + c_{\text{km}} D_R + \sum_{j \in R} [c_{\text{var},j} + (1 - \hat{p}_j) c_r + \mathbb{E}[\text{loss}_j]] \tag{3}$$

If the sum of allocated prices covers the sum of allocated costs, the route is provable. If not, the route fails regardless of how each drop is priced individually.

3 The allocation rule

The choice of s_j is the strategic decision. Four rules, in order of preference:

Rule	s_j	When to use
Equal	$1/N$	Drops are similar in size, time, and addressability
Time-weighted	$\mu_j / \sum_k \mu_k$	Drops differ in service time (addressability differs) — the def
Incremental distance	$\Delta d_j / D_R$	Drops are spread along the route and distance dominates
Ramsey (inverse elasticity)	$(1/\varepsilon_j) / \sum_k (1/\varepsilon_k)$	Drops have known, differing price sensitivities

The recommended default is time-weighted for the fixed component and incremental-distance for the distance component:

$$s_j = \alpha \cdot \frac{\mu_j}{\sum_k \mu_k} + (1 - \alpha) \cdot \frac{\Delta d_j}{D_R} \tag{4}$$

where $\alpha \in [0, 1]$ trades off time versus distance. For dense urban corridors, $\alpha = 0.7$; for long intercity routes, $\alpha = 0.3$.

Why time-weighted as the default: the driver's shift is the shared resource being consumed, and a drop that takes 20 minutes (bad address) consumes four times the shift of a drop that takes 5 minutes (good address). Pricing on time makes the customer with the badly-addressed endpoint bear the cost that endpoint imposes.

4 Worked example: CBD → Thika

The route: Depot (CBD) → Drop A (Kasarani, 12 km) → Drop B (Ruiru, 10 km further) → Drop C (Thika, 14 km further). Total: 36 km.

Inputs.

- Fixed costs: $c_0 = 1{,}300$ (rider fee), $\Phi C_p = 200$, $c_{\text{km}} = 25$/km
- Total route cost: $1{,}300 + 200 + 25(36) = 2{,}400$
- Addressability: A = 0.85, B = 0.50, C = 0.75
- Service time: A = 5 min, B = 20 min, C = 8 min (total 33 min)
- Incremental distance: A = 12 km, B = 10 km, C = 14 km
- All drops: $\hat{p} \approx 0.9$, $A(a) \approx 1.1$
- Standalone components (variable + failure + loss): A = 200, B = 350, C = 250

Step 1: Compute time-weighted shares ($\alpha = 0.7$).

Notice: B gets the largest share because B has the worst address. This is correct—B's unlocatable endpoint consumes 20 minutes of the driver's shift, so B should bear the largest share of the shared cost. This is the mechanism that makes bad addresses pay for themselves.

Step 2: Allocate shared cost and compute net prices.

Step 3: Route test.

Drop	Time share	Distance share	s_j (0.7×time + 0.3×distance)
A	$5/33 = 0.152$	$12/36 = 0.333$	$0.106 + 0.100 = 0.206$
B	$20/33 = 0.606$	$10/36 = 0.278$	$0.424 + 0.083 = 0.507$
C	$8/33 = 0.242$	$14/36 = 0.389$	$0.169 + 0.117 = 0.286$
Σ			$= 1.000$

Drop	Standalone cost	Allocated shared	n_j (net)	P_j (client price, VAT)
A	200	$0.206 \times 2{,}400 = 494$	$(200 + 494)/(0.9 \times 1.1) = 701$	813
B	350	$0.507 \times 2{,}400 = 1{,}217$	$(350 + 1{,}217)/(0.9 \times 1.1) = 1{,}582$	1,835
C	250	$0.286 \times 2{,}400 = 686$	$(250 + 686)/(0.9 \times 1.1) = 945$	1,096

The route passes, but by a thin margin (28 KES). This tells you the route is marginally viable and should be watched closely—if B's addressability were 0.40 instead of 0.50, the route would fail and B would need to be either re-priced upward or prepared (verified).

Step 4: Check the dedicated floor comparison.

This is the payoff of consolidation. Each customer pays less than a dedicated trip would cost them, and the route is still profitable. The savings are largest for the closest drop (A) because it benefits most from the route being extended beyond it.

5 The four operational rules

Rule 1: Never price a drop below its standalone component. The standalone component $c_{\text{var},j} + (1 - \hat{p}_j)c_r + \mathbb{E}[\text{loss}_j]$ is the cost the drop imposes regardless of whether it shares a route. If the allocated price does not cover this, the drop is being subsidised by the other drops on the route, and the route's economics are fragile.

Rule 2: Price the shared component by time share, not equally. A drop that consumes 20 minutes of the driver's shift should bear four times the shared cost of a drop that consumes 5. This is the mechanism that penalises bad addresses and rewards good ones—the same mechanism that drives the flywheel.

Rule 3: Offer a consolidation discount explicitly. Tell the customer: "If you can wait until 2pm when the Thika run departs, your price is KES 813. If you need it now, it's KES 1,350." This is the temporal lever from the consolidation theory, priced explicitly. The customer who accepts the discount is paying with time; the customer who refuses is paying with money. Both are valid choices.

Rule 4: Surcharge the marginal drop. If adding drop D to the route requires a detour of 15 km and delay of 25 minutes, the marginal cost is:

$$\Delta = c_{\text{km}} \cdot 15 + \mu_D \cdot w_{\text{driver}} + \Delta_{\text{cascade}} = 375 + \text{time cost} + \text{cascade} \tag{5}$$

Drop D must be priced to cover this marginal cost, or the route's contribution falls. The marginal drop test from the route optimization paper is the operational form of this rule.

6 Connection to the bucket system

Each route corresponds to a bucket (corridor × departure time). The bucket accumulates parcels until cutoff, then dispatches on a single route. The route pricing rule operates at two points:

1. At parcel arrival: the drop is assigned to a bucket, and the drop-level price (from PURM) is computed as the ceiling the customer will accept.

Line	KES
Total net revenue	$701 + 1{,}582 + 945 = 3{,}228$
Total allocated cost	$2{,}400 + (200 + 350 + 250) = 3{,}200$
Route contribution	$+28$

Drop	Shared price	Dedicated floor	Saving from consolidation
A	813	1,350	-537 (-40%)
B	1,835	2,180	-345 (-16%)
C	1,096	1,680	-584 (-35%)

2. At bucket cutoff: the actual route is computed, the allocation rule is applied, and the final prices are computed. If any drop's allocated price exceeds the customer's ceiling, the drop is prepared (held for the next bucket, at a lower price) or the route is repriced.

This is the operational form of the flywheel: the bucket accumulates density, the route allocates cost, the allocation prices each drop by its resource consumption, and the customer chooses between consolidation (lower price, longer wait) and priority (higher price, immediate dispatch).

7 The pricing formula in one line

Each drop pays its standalone cost plus a time-weighted share of the route's fixed and distance costs, divided by its success-and-addressability factor—and the route passes if the sum of allocated prices covers the sum of allocated costs, with the worst-addressed drop bearing the largest share because it consumes the most driver time.